Seunghun Lee

Postdoctoral Researcher, Machine Learning and Vision Lab, KAIST

llsshh319@kaist.ac.kr | Google Scholar

Research interests

Weather and climate forecasting with deep learning • Latent Bayesian optimization • Implicit neural representations • Graph neural networks

Education

Sep 2020 – Aug 2025

Ph.D., Computer Science and Engineering
Korea University, Seoul, Korea
Integrated M.S./Ph.D. program Machine Learning and Vision Lab
Advisor: Prof. Hyunwoo J. Kim

- Dissertation: *Latent Space Modeling for Latent Bayesian Optimization and Implicit Neural Representations*

Mar 2016 – Aug 2020

B.S., Computer Science and Engineering
Korea University, Seoul, Korea

Experience

Sep 2025 – Dec 2026

Postdoctoral Researcher
KAIST, Machine Learning and Vision Lab
LLM Innovation Research Center, InnoCORE Research Group, Ministry of Science and ICT

Projects

- **CLIMECAST: Extreme Climate Event Forecasting and Disaster Question-Answering via Multimodal Foundation Models and Latent Diffusion Models** (Jun 2024 – present). Project lead; funded by MSIT through the National Research Foundation of Korea. Developed continuous weather downscaling with neural fields and generative ODE models for atmospheric dynamics.
- **Virtual Engineering and Materials Data Platform for Digital Transformation of Ceramics** (Jan 2023 – present). Funded by MOTIE through KIAT, in collaboration with KICET. Developed Bayesian optimization methods for ceramic process parameters.

Sep 2020 – Aug 2025

Graduate Student Researcher
Korea University, Machine Learning and Vision Lab

Selected projects

- **Data-Driven Program Analysis** (Aug 2021 – Aug 2023). Funded by the Samsung Research Funding & Incubation Center of Samsung Electronics. Developed Bayesian optimization over learned latent spaces for de novo molecular design and structured black-box optimization, published at NeurIPS 2023.
- **A Hyper-Scale Multi-Purpose Graph Neural Network** (Oct 2021 – Sep 2022). Cooperative research with NAVER CLOVA MLX.
- **Spatio-Temporal Graph Neural Networks for Estimated Time of Arrival Prediction** (Jun 2021 – May 2022). Cooperative research with NAVER Maps. Developed spatio-temporal graph models and training schemes for travel-time estimation.
- **Effective Training Schemes for Graph Neural Networks and Applications in Recommender Systems** (Aug 2020 – Jul 2021). Cooperative research with NAVER CLOVA MLX. Developed graph data augmentation methods published at NeurIPS 2021 and in IEEE Access.

Publications

Asterisks denote equal contribution.

Under review

NeurIPS 2026

Learning to Solve Generative ODEs Beyond the Linear Span
Sihyeon Kim, **Seunghun Lee**, Vikas Singh, Hyunwoo J. Kim
NeurIPS 2026, under review. arXiv:2606.08672

In preparation

2026

Climatology-Based Residual Correction with Neural Fields for Continuous Weather Downscaling
Seunghun Lee, Sihyeon Kim, Injae Kim, Minseo Yoon, Hyunwoo J. Kim
Manuscript in preparation.

Conference papers

NeurIPS 2025

PRESTO: Preimage-Informed Instruction Optimization for Prompting Black-Box LLMs
Jaewon Chu, **Seunghun Lee**, Hyunwoo J. Kim
Advances in Neural Information Processing Systems, 2025.

ICLR 2025

Latent Bayesian Optimization via Autoregressive Normalizing Flows
Seunghun Lee, Jinyoung Park, Jaewon Chu, Minseo Yoon, Hyunwoo J. Kim
International Conference on Learning Representations, 2025. Oral presentation, top 1.8% of submissions.

NeurIPS 2024

Inversion-based Latent Bayesian Optimization
Jaewon Chu*, Jinyoung Park*, **Seunghun Lee**, Hyunwoo J. Kim
Advances in Neural Information Processing Systems, 2024.

NeurIPS 2023

Advancing Bayesian Optimization via Learning Correlated Latent Space
Seunghun Lee*, Jaewon Chu*, Sihyeon Kim*, Juyeon Ko, Hyunwoo J. Kim
Advances in Neural Information Processing Systems, 2023.

NeurIPS 2023

NuTrea: Neural Tree Search for Context-guided Multi-hop KGQA
Hyeong Kyu Choi, **Seunghun Lee**, Jaewon Chu, Hyunwoo J. Kim
Advances in Neural Information Processing Systems, 2023.

NeurIPS 2021

Metropolis-Hastings Data Augmentation for Graph Neural Networks
Hyeonjin Park*, **Seunghun Lee***, Sihyeon Kim, Jinyoung Park, Jisu Jeong, Kyung-Min Kim, Jung-Woo Ha, Hyunwoo J. Kim
Advances in Neural Information Processing Systems, 2021.

Journal papers

NN 2022

Graph Transformer Networks: Learning Meta-path Graphs to Improve GNNs
Seongjun Yun, Minbyul Jeong, Sungdong Yoo, **Seunghun Lee**, Sean S. Yi, Raehyun Kim, Jaewoo Kang, Hyunwoo J. Kim
Neural Networks, 2022.

IEEE Access 2021

Learning Augmentation for GNNs with Consistency Regularization
Hyeonjin Park*, **Seunghun Lee***, Dasol Hwang, Jisu Jeong, Kyung-Min Kim, Jung-Woo Ha, Hyunwoo J. Kim
IEEE Access, 2021.

KIISE 2021

Hyperbolic Graph Transformer Networks for non-Euclidean Data Analysis on Heterogeneous Graph
Seunghun Lee, Hyeonjin Park, Hyunwoo J. Kim
Journal of KIISE, 2021.

Patents & software

P1 **Deep Learning Model-Based Optical Character Recognition Method for Indoor Place Name Prediction and Apparatus Therefor**
Hyunwoo J. Kim, Sojin Lee, Minseok Joo, **Seunghun Lee**
Korean Patent 10-2613984, registered Dec 11, 2023.

C1 **WINDY: Weather Forecasting Based on Implicit Neural Representations and Neural ODEs**
Hyunwoo J. Kim, **Seunghun Lee**
Software copyright C-2025-051886, registered to KAIST with the Korea Copyright Commission, Nov 24, 2025.

C2 **Graph-Based Probabilistic Weather Forecasting with Latent Diffusion Models**
Hyunwoo J. Kim, Minseo Yoon, Jinyoung Park, **Seunghun Lee**, Minsung Bae, Jaewon Chu
Software copyright C-2025-046973, registered to KAIST with the Korea Copyright Commission, Nov 5, 2025.

Honors & awards

- Honorable Mention, 4th Weather-AI Boostcamp, National Institute of Meteorological Sciences, Korea Meteorological Administration, 2025. Awarded at the WMO AI-NPP Workshop.

Grants & funding

2026 **Cross-Domain Super-Resolution Weather Forecasting for the Korean Peninsula**
Principal investigator

- Computing allocation awarded through the 2026 National AI Data Center program, Artificial Intelligence Industry Cluster Agency (AICA): NVIDIA H100 GPUs and 10 TB of storage, from May 2026.

Academic service

Reviewer

- AAAI, 2027
- NeurIPS, 2026
- ACCV, 2026
- MedAGI, MICCAI Workshop on Foundation Models for General Medical AI, 2024 and 2026